A Macroeconomic Approach to Optimal Fertility Across Nations

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October 22, 2025

Preliminary Abstract: The latest version is available here

Abstract

Should countries aim for the replacement level of fertility? In other words, is the optimal fertility rate below or above the replacement level? Does the answer differ across countries, and if so, why? This project explores these questions through the lens of a dynastic model that captures two opposing channels through which fertility affects the long-run national income and welfare. On the one hand, higher fertility generates positive externalities, as a larger population contributes to a greater stock of ideas and a larger labor force in the long run. On the other hand, negative (fiscal) externalities arise from fertility through the child penalty and the quantity–quality trade-off, both of which reduce aggregate labor input and raise the equilibrium tax rate. Using data from nine high-income countries, I calibrate and estimate key parameters governing preferences, technology, and government spending to match cross-country differences in fertility, education, leisure, public expenditure, and income levels. Solving the planner's problem for each country, I find that although the optimal fertility rates exceed the equilibrium rates, they still remain below the replacement level in many countries. This is particularly true for countries such as Germany, Japan, and Spain, where higher child penalties raise the social costs of increased fertility through the associated output losses. These preliminary results suggest that, without measures to reduce the social costs of higher fertility—such as the child penalty—the rationale for pronatal policy weakens considerably.

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